RL-Tuned Synthetic Marine Radar for Sim-to-Real Transfer

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***Abstract*—Machine-learning models for marine radar are often trained on synthetic data because real labeled logs are expensive to collect. A key challenge is the sim-to-real gap: models that perform well in simulation can fail on real sensor data. This proposal studies an environment-aware reinforcement learning controller that tunes simulator parameters using feedback from held-out real validation data, with environment labels such as lake, river, and coast. We compare the method against fixed randomization, Bayesian optimization, and unconditioned RL baselines, and evaluate both performance gains and sample efficiency under domain holdout and stress conditions.**

I. Introduction

Marine radar is useful for obstacle detection, situational awareness, and local mapping, but robust learning requires large labeled datasets. Because collecting and labeling maritime radar logs is expensive, many pipelines rely on synthetic data. This creates a persistent sim-to-real gap: models that perform well in simulation can degrade substantially on real sensor data.

This proposal asks whether simulator parameters can be tuned automatically so synthetic training distributions better match real maritime operating regimes. We study an RL controller that updates simulator parameters to maximize held-out real-validation performance. The controller is conditioned on broad environment type ('lake', 'river', 'coast') with the hypothesis that context-aware tuning outperforms fixed settings and uniform randomization, especially under route and clutter shift.

The method builds on domain-randomization and adaptive sim-to-real literature [1]–[9]. Radar transfer motivation is further supported by recent radar-domain studies [10], [11] and by public real-radar datasets that expose weather and domain shift challenges [12]–[16]. Prior work often uses fixed/random perturbations, one-shot adaptation, or calibration pipelines without explicit maritime context. Our contribution is an environment-aware RL outer loop that optimizes downstream real-task reward directly (Fig. **??**) and is compared against explicit capability gaps in related work (Fig. **??**).

II. Approach

We cast simulator tuning as a bi-level optimization problem. Let $\mathcal{M}$ be the downstream perception model (e.g., MRIS-Net for ship segmentation [17]) and $\mathcal{S}(\theta, e)$ be a physically grounded radar simulator parameterized by θ and conditioned on environment type $e \in \{\text{lake}, \text{river}, \text{coast}\}$. The objective is

$$\theta^* = \arg\max_{\theta} \mathcal{R}\left(\mathcal{M}^*(\theta), \mathcal{D}_{real}\right) \tag{1}$$

subject to inner-loop training

$$\mathcal{M}^*(\theta) = \arg\min_{\omega} \mathcal{L}_{task}\left(\mathcal{M}_{\omega}, \mathcal{S}(\theta, e)\right). \tag{2}$$

The outer-loop policy $\pi(\theta_{t+1} \mid s_t)$ proposes sequential updates to simulator parameters. State s_t includes current θ, environment label e, recent validation metrics, and trajectory history. Reward is

$$r_t = \Delta \text{mIoU}_{real} - \lambda_1 \|\theta_t - \theta_{init}\|_2 - \lambda_2 \mathbb{1}(\theta \notin \Omega_{phys}), \tag{3}$$

where Ω_{phys} denotes physically plausible bounds.

The simulator models multipath and clutter using physically meaningful factors: speckle scale, attenuation, clutter floor, and multipath gain. To avoid unrealistic exploitation, each action is clipped to calibrated parameter ranges and penalized when violating physical plausibility. We use Soft Actor-Critic (SAC) for the outer loop because it supports stable learning in continuous action spaces with stochastic exploration.

The key design difference from prior adaptive randomization is that this policy is (i) sequential, (ii) environment-conditioned, and (iii) optimized on downstream real-task reward rather than simulator similarity alone.

III. Evaluation

Evaluation uses real radar and maritime benchmarks that expose domain shift and adverse conditions, including PoLaRIS and LaRS, with auxiliary stress testing on radar-focused datasets such as RADIATE and CARRADA [12]–[16]. We use strict splits by route/location and leave-one-environment-out testing to prevent leakage.

We compare five settings: (1) fixed simulator defaults, (2) uniform random domain randomization, (3) Bayesian optimization tuning, (4) RL tuning without environment labels, and (5) RL tuning with environment labels (proposed).

Primary outcomes are real-data task performance and sim-to-real gap reduction. Secondary outcomes include cross-domain holdout performance, unseen-route robustness, run-to-run variance, and low-SNR/high-clutter stress behavior. We also report real-validation-call efficiency to quantify adaptation cost.

Each novelty claim is tied to a direct test: (a) sequential RL adaptation versus BayRn under matched real-evaluation budget [5]; (b) environment-conditioned versus unconditioned RL; and (c) realism-constrained versus unconstrained action policies.

Published evidence already motivates this direction (Fig. **??**). Real FMCW transfer can degrade toward chance-level behavior without adaptation, while calibrated adaptation improves substantially [11]. In synthetic-to-measured SAR transfer, stronger structured adaptation outperforms synthetic-only and simple-noise baselines [10]. These patterns motivate feedback-driven simulator control instead of fixed randomization.

IV. Conclusion

This project targets a practical problem with broad relevance beyond radar: models trained in simulation often fail when deployed in the real world. We propose an environment-aware RL controller that tunes synthetic data generation to improve real-data task performance, and we benchmark it against strong non-RL alternatives under realistic transfer stress tests. If full RL tuning is unstable, a contextual-bandit or Bayesian fallback still yields a complete and informative outcome.

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